STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 10A

Description

10 a) Write R program to find Time Series Analysis with the sample data and visualize the regression graphically.

Procedure and Overvation:

Description:

TIME SERIES ANALYSIS:

It is a series of data points in which each data point is associated with a time stamp.

ex:1)Price of a stock in the stock market at the different points of a time on a given day.

2)Amount of rainfall in a region at different months of the year.

R language uses many functions to create, manipulate and plot the timeseries data. The data for the timeseries is stored in ana R object called timeseries object. It is also an R data object like a vector (or)data frame.

SYNTAX:

ts()

timeseries object name<-ts (data, start, end, frequency)

where

data is a vector or matrix containing the values used in time series

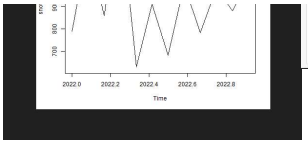
start specifies the start time for the first observation in timeseries

end specifies the end time for the last observations in time series.

Frequency specifies the number of observations for unit time.

SOURCE CODE:

```
#Getting the data points in form of a R vector.
snowfall<-c(790,1170.8,860.1,1330.6,630.4,911.5,683.5,996.6,783.2,982,881.8,1021)
#Converting it into a time series object.
snowfall_timeseries<-ts(snowfall,start=c(2022,1),frequency=12)
#Printing the time series data.
print(snowfall_timeseries)
#Giving a name to the chart file.
png(file="snowfall.png")
#Plotting a graph of the time series.
plot(snowfall_timeseries)
#Saving the file.
dev.off()
```



VIVA RESPONSES

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